Arista Access Node WFD-ACC-48 Standard Operating Procedure (SOP) for Routing Protocol Flap (ALARM_6376)

Node: WFD-ACC-48 | **Vendor:** Arista | **Severity:** Warning | **Alarm Code:** ALARM_6376

Description

This Standard Operating Procedure (SOP) addresses the occurrence of Routing Protocol Flap (BGP route churn, OSPF adjacency loss, ISIS LSP flooding) on Arista Access Node WFD-ACC-48 at the Whitefield POP. The Root Cause Analysis (RCA) reveals that this failure domain arises primarily due to the dynamic nature of routing protocols, which inherently induce frequent routing updates, causing instability in the routing table. This instability can be exacerbated by configuration errors, misaligned routing policies, or suboptimal network designs. Furthermore, the use of OSPF and ISIS in conjunction with BGP may introduce additional complexities, such as route oscillations, which can lead to adjacency loss and LSP flooding. The Arista equipment's robustness and scalability are expected to mitigate these issues, but it is essential to promptly address any anomalies to maintain network stability.

Pre-requisite Checks

- Check: Verify the power and cooling systems at the Whitefield POP are functioning within normal operating parameters.

- Check: Confirm the redundant link status, ensuring that alternative paths are available for traffic diversion.

Potential Root Causes

Cause	Probability
BGP Route Churn	High
OSPF Adjacency Loss	Medium
ISIS LSP Flooding	Low

Diagnostic Commands & Expected Outputs

```
Command: show ip bgp, Expected Output: BGP table version is 1, main routing table version 1 Routing Table Entries: 2 UnBESTPATH: 0, Description: This command displays the BGP routing table and version information, helping identify potential route churn and instability.

Command: show ospf neighbor, Expected Output: Neighbor ID Pri State Dead Time Interface FLG RTRID BDRID NbrIndex , Description: This command displays OSPF neighbor information, which can help identify adjacency loss and potential configuration errors.
```

```
Command: show isis database, Expected Output: ISIS LSP Database: LSP ID Seq# Age Metric Next-Hop IP Address , Description: This command displays the ISIS LSP database, aiding in the identification of LSP flooding and potential network design issues.
```

Resolution Steps

Step 1: Step: Verify and Correct Configuration, Command: show running-config | i router bgp, Description: Inspect the BGP configuration for any errors or inconsistencies, and correct them as necessary to prevent route churn.

Step 2: Step: Review Routing Policies, Command: show policy-map interface, Description: Verify that routing policies are correctly configured and not contributing to route oscillations or instability.

Step 3: Step: Optimize Network Design, Command: show cdp neighbors, Description: Review the network topology and adjust as necessary to minimize potential issues with OSPF and ISIS adjacency loss and LSP flooding.

Post-Check Verification

```
Command: show ip bgp, Expected Output: BGP table version is 2, main routing table version 2 Routing Table Entries: 3 UnBESTPATH: 0, Description: Verify that the BGP routing table has stabilized and version information has incremented.

Command: show ospf neighbor, Expected Output: Neighbor ID Pri State Dead Time Interface FLG RTRID BDRID NbrIndex , Description: Verify that OSPF adjacencies have been restored and potential configuration errors have been addressed.

Command: show isis database, Expected Output: ISIS LSP Database: LSP ID Seq# Age Metric Next-Hop IP Address , Description: Verify that ISIS LSP flooding has subsided and network design issues have been resolved.
```

Roll-back Procedures

- Step: Restore Previous Configuration, Command: copy startup-config running-config, Description: Rollback to the previous configuration to restore the network to its previous state in case the fix fails.

- Step: Re-apply Previous Fixes, Command: copy startup-config running-config, Description: Re-apply the previous fixes or corrections to the BGP, OSPF, and ISIS configurations to maintain network stability.